Wraysbury Materials & Engineering Laboratory

Forensic Engineering Report — Guard Interlock Failure Analysis

Prepared for Whitfield Carter LLP, for the dominant purpose of anticipated proceedings — PRIVILEGED

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Subject equipment	Excelon EX-450 thread-rolling cell, asset CR-TR-12
Instructing solicitors	Whitfield Carter LLP

1. Instructions and scope

I was instructed to examine thread-rolling cell CR-TR-12 and to determine (a) the means by which the guard interlock came to be defeated; (b) whether the interlock arrangement met the requirements of BS EN ISO 14119 for the prevention of defeat in a reasonably foreseeable manner; and (c) whether the defeat would have been apparent to operators, supervisors or maintenance staff.

2. Examination

The cell guard is secured by a single tongue-and-slot interlock switch of an older type that accepts a separate, removable actuator. I established that a spare actuator, held loose in the maintenance store, could be inserted into the switch to hold the interlock in the closed (safe) state with the guard door open. Insertion takes a few seconds and requires no tools. Once inserted, the machine could be run with the guard open. The machine-control event log confirms the interlock circuit was held closed continuously from 09:48 on 18 March 2026 until the emergency stop at 14:20.

3. Assessment against standards

The interlock arrangement does not meet the current expectation in BS EN ISO 14119 that interlocking devices be designed or installed so as to minimise the possibility of defeat in a reasonably foreseeable manner. The use of a single, coded but readily duplicated actuator that can be held by a freely available spare is a recognised weakness. Modern good practice (and the OEM's current specification) uses trapped-key or RFID-coded interlocks with dual-channel monitoring, which would have prevented the defeat observed here.

4. Foreseeability

In my opinion the defeat of this style of interlock by means of a spare actuator is a well-known and reasonably foreseeable failure mode. The presence of an uncontrolled spare actuator in the maintenance store, combined with production pressure to clear mis-rolls quickly, made such defeat likely over time. The condition would have been detectable on a competent inspection of the guarding arrangement.

5. Conclusions

The injury resulted from access to the rotating dies through a guard whose interlock had been defeated using a spare actuator. The interlock design was not adequate to prevent reasonably foreseeable defeat, and the spare actuators were not controlled. Retrofitting trapped-key interlocking with dual-channel guard monitoring, together with secure control of actuator keys, would address the failure mode.

Dr Faisal Iqbal — Wraysbury Materials & Engineering Laboratory — 16 April 2026